
Context-aware evaluation for fair and actionable store decisions

Store Performance Scoring

The Problem: Unfair Store Comparison

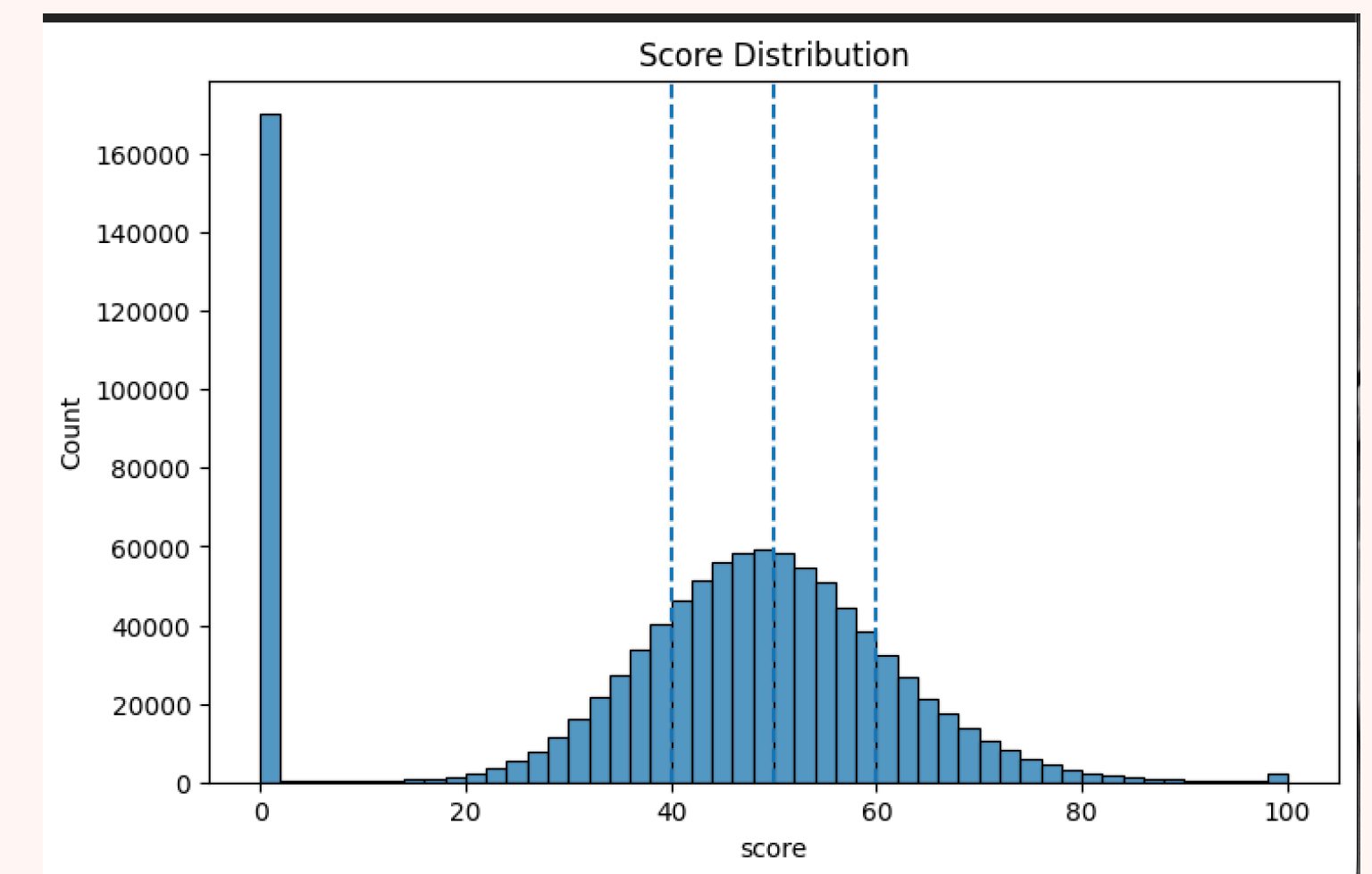
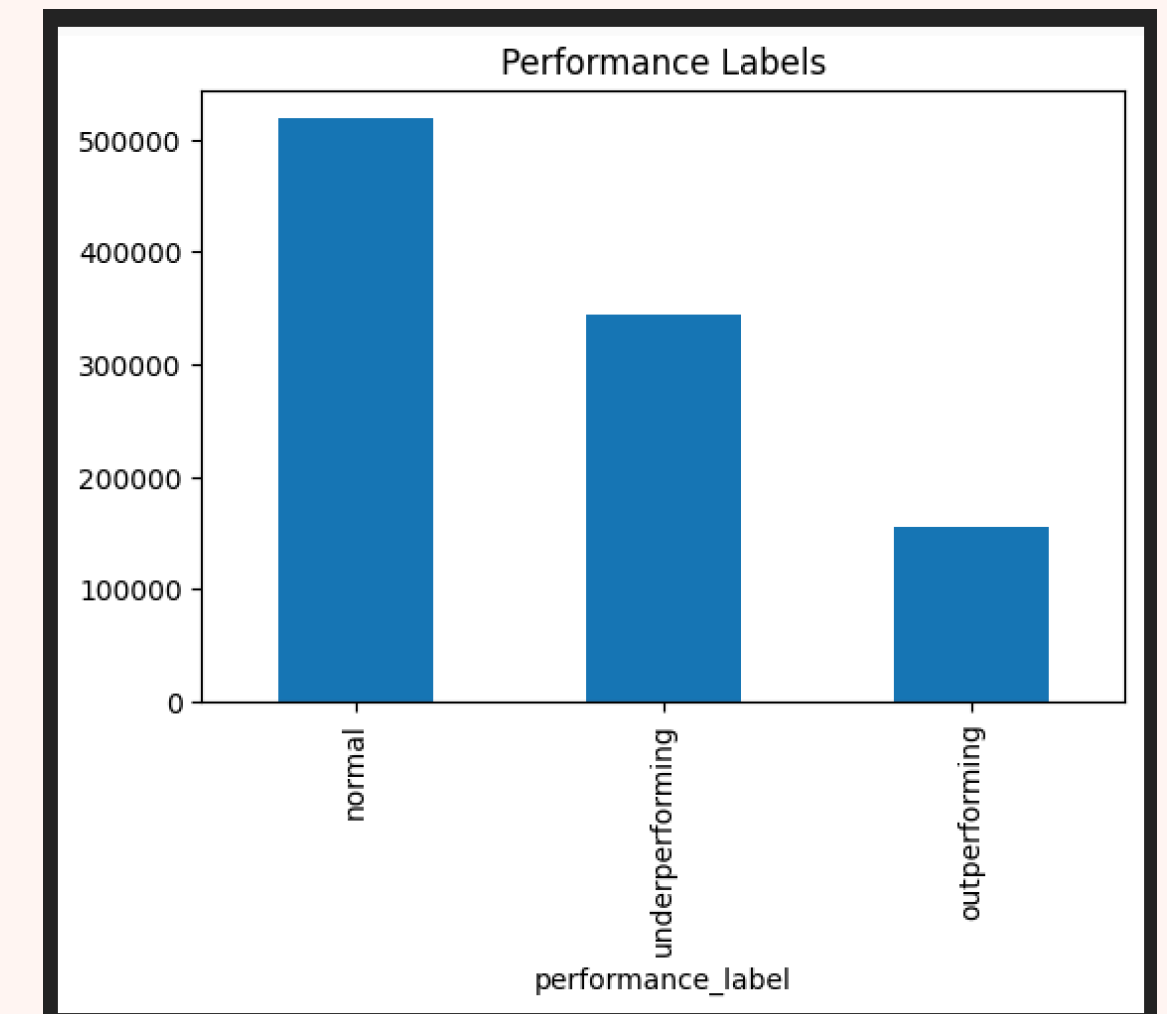
- **Store performance is commonly evaluated using raw KPIs such as sales or conversion rate**
 - **However, stores operate under significantly different conditions:**
 - ◆ **Customer traffic (footfall)**
 - ◆ **Promotions and markdown strategies**
 - ◆ **Store type and location**
 - **Because of these differences, direct comparison between stores is often misleading**
 - ✓ **The main challenge is to distinguish true performance from external effects**
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Approach

- **The idea is to estimate what each store is expected to achieve**
 - **Expected sales are predicted using store context and conditions**
 - **Actual sales are compared against expected values**
 - **The difference is normalized and transformed into a score**
 - ✓ **This allows performance to be evaluated relative to each store's environment**
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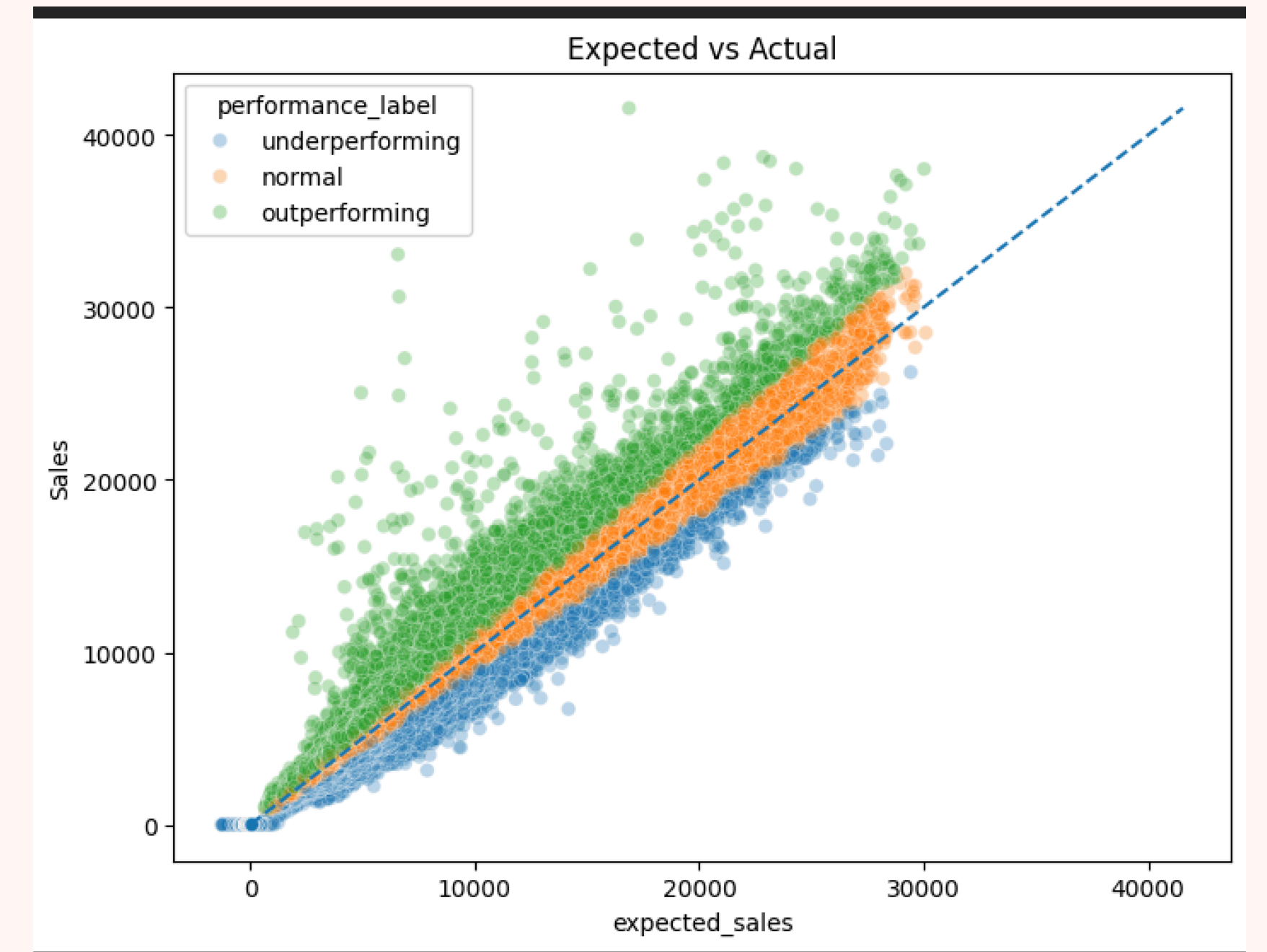
Model Output and Score Behavior

- The scoring system produces a continuous performance score
- Most stores cluster around expected performance (~50)
- Stores are grouped into:
 - ◆ Underperforming
 - ◆ Normal
 - ◆ Outperforming
- ✓ The model provides a stable baseline for comparing stores



Expected vs Actual Sales

- **The model estimates expected sales based on contextual features**
- **The diagonal line represents perfect prediction**
- **Points above the line indicate overperformance**
- **Points below the line indicate underperformance**
- ✓ **The gap between expected and actual sales is the core signal for scoring**



Example: Store-Level Analysis (Best vs Worst Store)

🔴 Underperforming Store (Store 425)

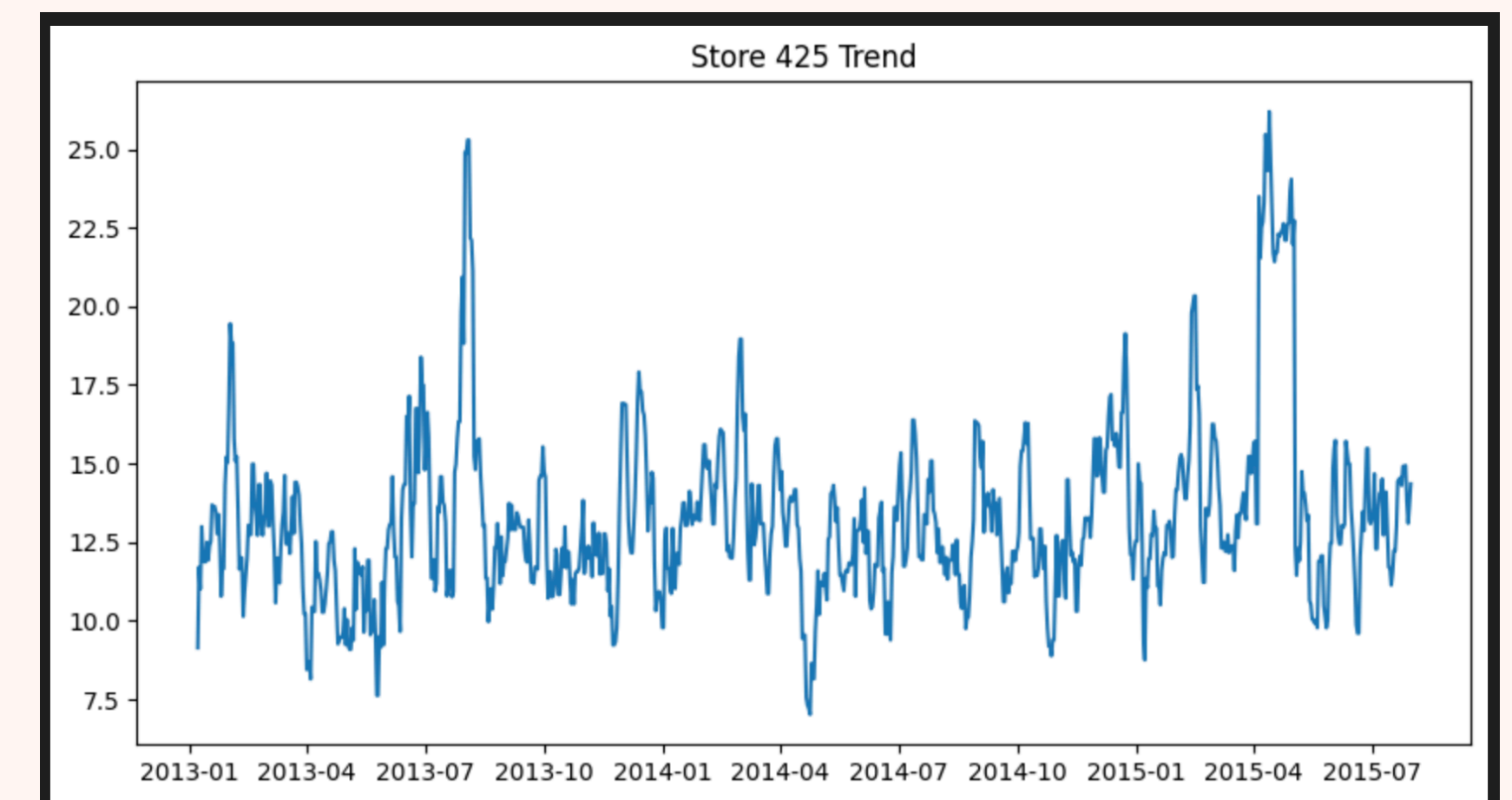
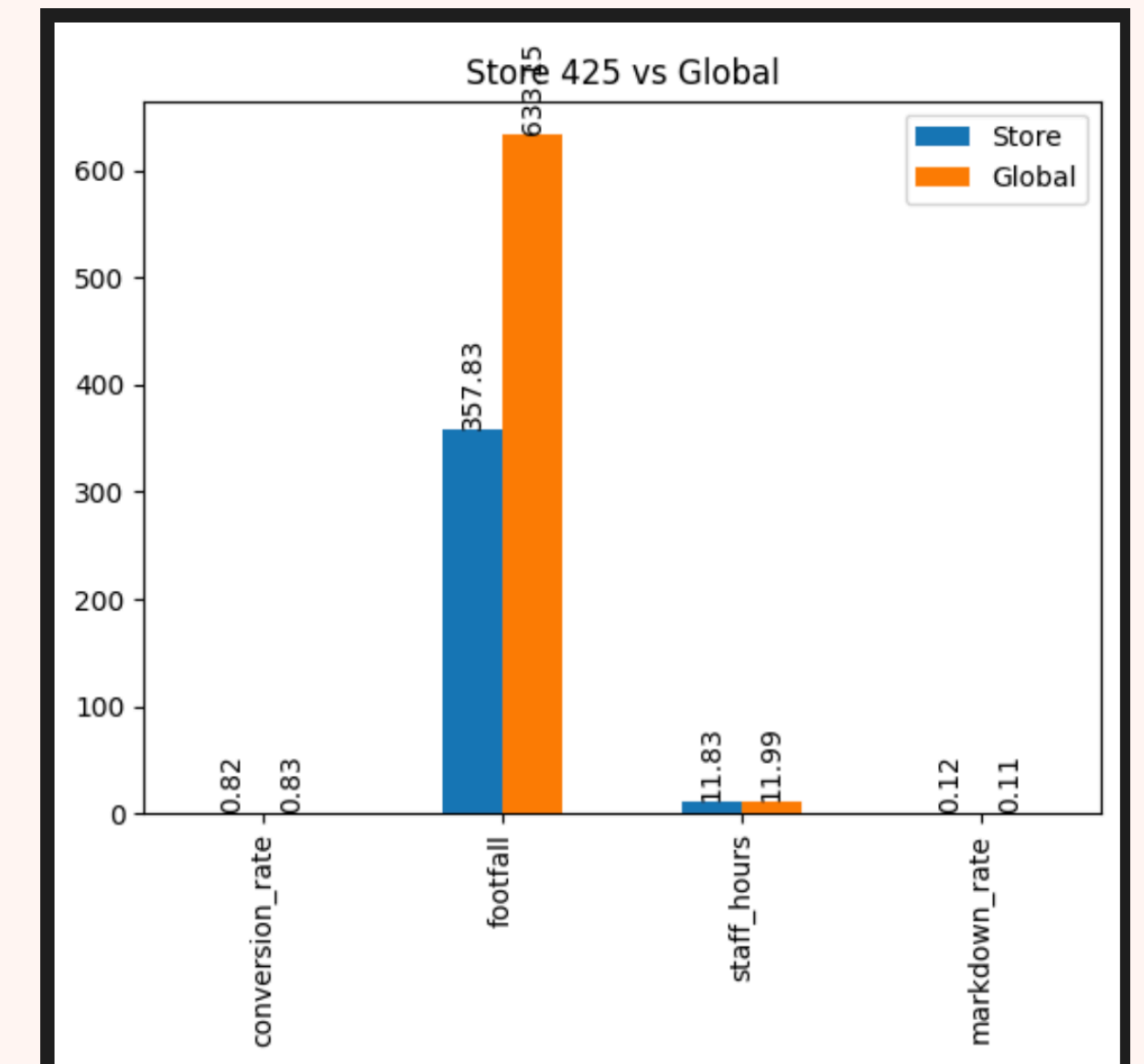
- ✦ Conversion rate is slightly below global (0.821 vs 0.829)
- ✦ Footfall is significantly lower than global (358 vs 633)
- ✦ Staff hours are aligned with global levels (~12)
- ✦ Markdown rate is similar to global (~0.113 vs 0.114)

👉 Interpretation:

- ✓ The store suffers primarily from low demand (footfall), not operational issues
- ✓ Even when customers visit, conversion is only slightly below average

👉 Conclusion:

- ✓ Performance gap is mainly driven by lack of traffic, not store execution



Example: Store-Level Analysis (Best vs Worst Store)

🟢 Outperforming Store (Store 139)

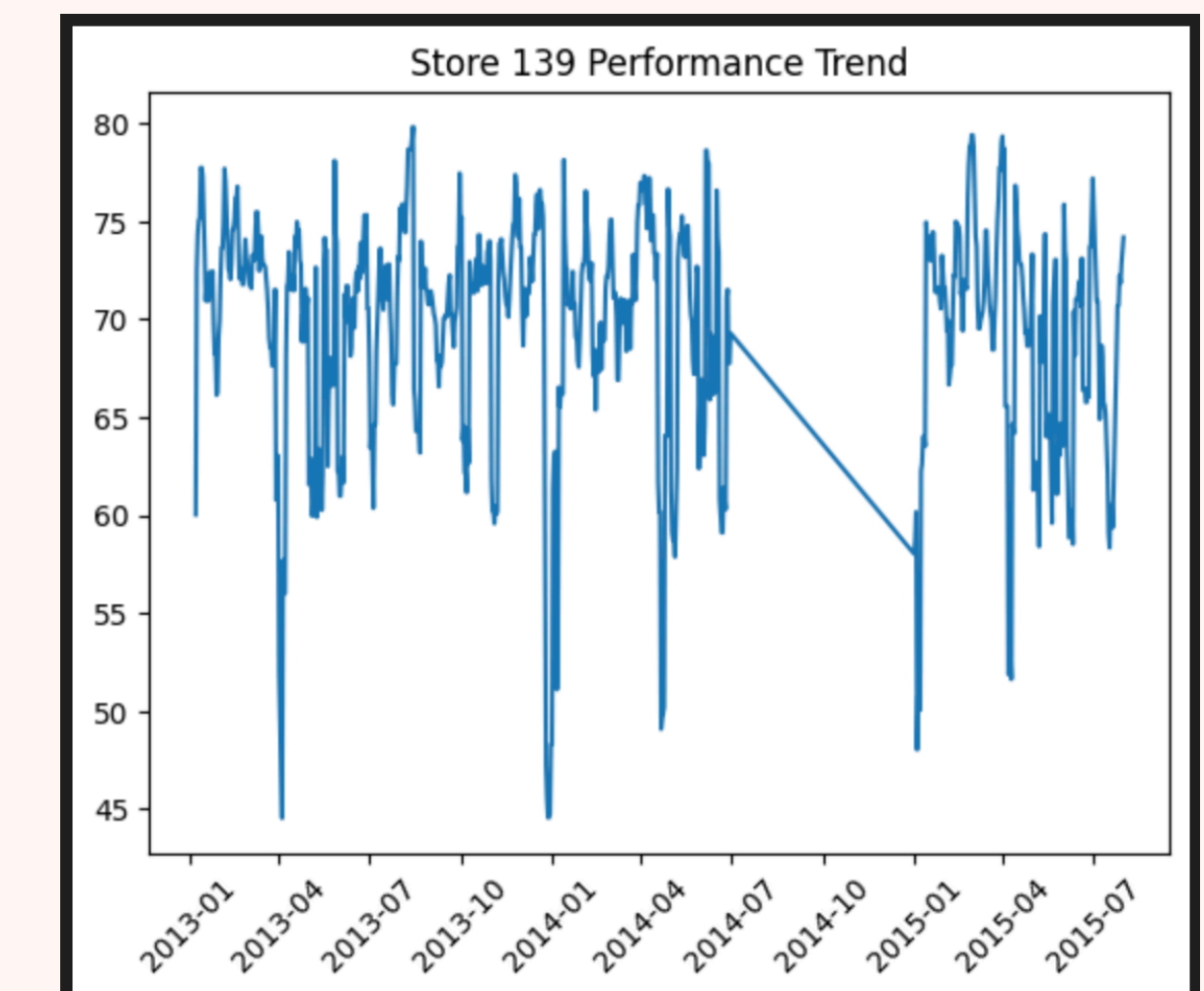
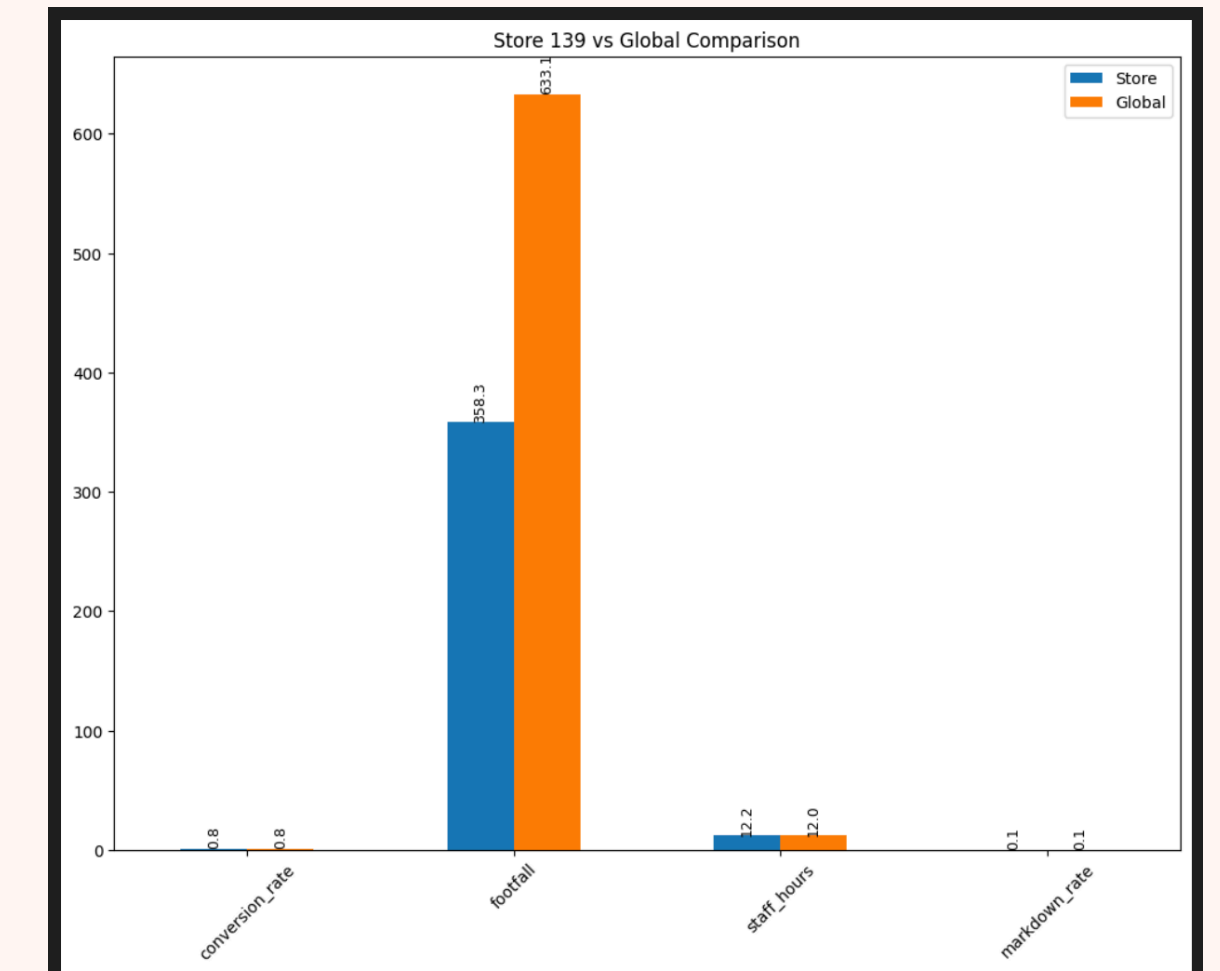
- ✦ Conversion rate is also below global (0.819 vs 0.829)
- ✦ Footfall is lower than global but higher than worst store (421 vs 358)
- ✦ Staff hours are consistent with global (~12)
- ✦ Markdown rate is slightly lower than global

👉 Interpretation:

- ✓ Despite operating under similar constraints (low traffic, similar staffing),
- ✓ This store achieves better performance than peers

👉 Conclusion:

- ✓ This indicates stronger execution or local advantages not captured in the model



Limitations and Improvements

CONTENT:

- ✦ **Some operational variables are approximated (e.g. footfall, staffing)**
- ✦ **The model does not explicitly capture temporal dependencies**
- ✦ **No hierarchical structure (region → country) is applied**

Next steps:

- ✦ **Use real operational data**
- ✦ **Introduce time-series modeling**
- ✦ **Add hierarchical or causal analysis**

👉 Closing:

- ✓ **The current solution provides a solid baseline that can be extended further**

THANK YOU
